Virtual University	CS609-System Programming (Solved Macq's) For Final Term CURRENT SUBJECTIVE	VU LMS HELP vulmshelp.com
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1. **How a DLL file is loaded inside a program during explicit linking? Mention the API that is used for this purpose.**
   ANSWER:

   Explicit linking requires the program to explicitly a specific DLL to be loaded or freed. Once the DLL is loaded then the program obtains the address of the specific entry point and uses that address as a pointer in function call. The function is not declared in the calling program. Rather a pointer to a function is declared. The functions required are:

   HMODULE **LoadLibrary**(LPCTSTR lpLibFileName);

2. **Write the names of functions that are used to determine the amount of time consumed by:**
   1. Process
   2. Thread

   ANSWER:

   - GetProcessTime()
   - GetThreadTime()

3. **When a thread enters into the critical section and is not willing to leave the critical section, then what will its consequences?**
   ANSWER:
   If a thread enters into the critical section and is not willing to leave the critical section, the following consequences occurs.
   - Deadlock
   - Starvation
   - Reduce System Performance

4. **Write the first two steps required for mapping a file to a region in virtual address space of a calling process.**
   ANSWER:
   - Open File with at least GENERIC_READ access.
   - Map the file using CreateFileMapping() or OpenFileMapping ()

5. **What is the purpose of InitializeCriticalSection and Spin Count API? Also mention the names of its parameter.**
   ANSWER:

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The InitializeCriticalSectionAndSpinCount API is used to initialize a critical section object in Windows with a spin count, which allows for better performance by using spinning instead of waiting. Its parameters are the critical section object to initialize and the spin count.

The names of its parameter:

VOID InitializeCriticalSection(LPCRITICAL_SECTION lpCriticalSection)

6. **What is Explicit Linking and which Windows API is used to load a DLL explicit inside a program?**

ANSWER

Explicit linking requires the program to explicitly a specific DLL to be loaded or freed. Once the DLL is loaded then the program obtains the address of the specific entry point and uses that address as a pointer in function call. The function is not declared in the calling program. Rather a pointer to a function is declared. The functions required are:

HMODULE **LoadLibrary**(LPCTSTR lpLibFileName);

The Windows API used to load a DLL explicitly inside a program is called "LoadLibrary".

7. **Name any Two Windows API That are used to return the Pseudo and real handle of a process.**

ANSWER:

- GetCurrentProress()

  GetCurrentProcess() is used to create a psuedohandle. It's not inheritable and desired access attributes are undefined.
- GetCurrentProcessId()

  The OpenProcess() function is used to obtain the process handle using the process ID obtained by GetCurrentProcessId() or otherwise.

8. **Briefly Describe the two parameters associated with the following windows API. FARPROC GetProcessAddress(HMODULE hModule, LPCSTR lpProcName).**

ANSWER:

The Windows API "GetProcAddress" is used to retrieve the address of an exported function or variable from a specified dynamic-link library (DLL).

It takes two parameters

- **HMODULE hModule:** is an instance produced by LoadLibrary
- **LPCSTR lpProcName** lpProcName cannot b UNICODE. It is the name of the function whose entry point is to be obtained. Its return

type is FARPROC.

9. **In what situation the termination handler is ignore by a process**
   ANSWER:

   A process can exit using the API ExitProcess(). The function does not return, in fact the process and all its associated threads terminate. Termination Handlers are ignored.

10. **Which window function is used to convert thread into fiber?**
    ANSWER:
    ConvertThreadToFiber

11. **Write the signature of InterlockedExchange() API with its return type and parameters**
    ANSWER:

```
LONG InterlockedExchange(
  LONG volatile *Target,
  LONG        Value
);
```

    return type and parameters is Loong Integer

12. **Write the name of the API which is used to map the view of a file mapping into the address space of a process**
    ANSWER:

```
LPVOID MapViewOfFile( HANDLE hFileMappingObject,
                      DWORD dwDesiredAccess,
                      DWORD dwFileOffsetHigh,
                      DWORD dwFileOffsetLow,
                      SIZE_T dwNumberOfBytesToMap );
```

Note: The name of the API is **MapViewOfFile**.

13. **Write the name of the API which is used to open map the view of a file mapping into the address space of a process?**

    ANSWER:

```
HANDLE OpenFileMapping(
                      DWORD dwDesiredAccess,
                      BOOL bInheritHandle,
                      LPCSTR lpMapName );
```

Note: The name of the API is **openFileMapping**.

14. **Suppose we have a semaphore named as hThrottleSem objects current count is to be increased by 1 and the previous count is not required. you need to write the code to release this semaphore using the above-mentioned information.**
Note: Make use of Windows API to release semaphore.
ANSWER:

ReleaseSemaphore function. code snippet to release the hThrottleSem semaphore and increase itscount by 1:

ReleaseSemaphore(hThrottleSem, 1, NULL);

OR

```
#include <windows.h>

// Assuming hThrottleSem is a global or shared handle to the semaphore object

BOOL bPreviousCountNotNeeded = TRUE; // Set to TRUE to indicate previous count is not required

LONG lReleaseCount = 1; // Increase semaphore count by 1

LONG* lpPreviousCount = (bPreviousCountNotNeeded ? NULL : &lReleaseCount); // Pass NULL if previous count is not needed

BOOL bSuccess = ReleaseSemaphore(hT\hrottleSem, lReleaseCount, lpPreviousCount);

if (!bSuccess) {
   // Handle error
}
```

15. **Pointer declaration Windows: LPTSTR pFile = Null; define two based pointers on the above pointer. take pSize and pIn as their names. their types should be DWORD and TCHAR respectively.**
ANSWER:

Assuming that TCHAR is defined as a wide-character (Unicode) character type, here's how you candefine two base pointers pSize and pIn based on the LPTSTR pointer pFile:

Code:

```
DWORD* pSize = nullptr;
```

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```
TCHAR** pIn = nullptr;

pSize = reinterpret_cast<DWORD*>(pFile);
pIn = reinterpret_cast<TCHAR**>(pFile + sizeof(DWORD));
```

Here, pSize is a pointer to a DWORD, and pIn is a pointer to a pointer to a TCHAR.

16. What is the purpose of *Addend and Value parameter used in InterlockedExchangeAdd() API?

ANSWER:

Increment is added to the *addend and the original value of *addend is returned.

```
LONG InterlockedExchangeAdd(
LONG volatile *Addend,
LONG Increment);
```

17. VOID sleep(DWORD dwMilliseconds);

i. **In above API if the parameter dwMilliseconds is specified as INFINITE then what happened with threads.**

ii. **If the parameter dwMilliseconds is specified as 0 then what happened with threads?**

ANSWER:

dwMilliseconds is the time period for wait

- If the dwMilliseconds parameter is specified as INFINITE:
  INFINITE for indefinite wait
- If the dwMilliseconds parameter is specified as 0:
  0 for no wait

18. (Give answer in Yes No)

I. **Is it incorrect that Actual handle has its own security and access attributes while the Pseudo handle has, no?**

II. **Name the API used for getting Pseudo handle.**

III. **Name the API used for getting Actual handle.**

ANSWER:

I. Yes, it is incorrect. Both actual handles and pseudo handles have security and access attributes associated with them, which are used to determine whether a process or thread has permission to access the object represented by the handle.

II. The API used for getting a pseudo handle is GetCurrentProcess.

III. The API used for getting an actual handle depends on the type of object being opened.

OR

The API used for getting an Actual handle is process ID

19. Write the names of 4 functions used to create and manage Semaphores.

ANSWER:

- CreateSemaphore()
- CreateSemaphoreEx()
- OpenSemaphore()
- ReleaseSemaphore().

20. Briefly describe the following parameters that are used with CreateProcess0 API.

I. **LPCSIR lpApplication Name**
II. **BOOL bInheritHandles**
III. **LPVOID lpEnvironment**
IV. **LPCSIR 1pCurrentDirectory**

ANSWER:

- **LPCSTR lpApplicationName:** It specify the program name
- **BOOL bInheritHandles**: It specifies weather the new inherit copies of the calling process inherited handle.
- **LPVOID lpEnvironment:** This parameter is a pointer to an environment block for the new process. If this parameter is NULL, the process uses parent's process.
- **LPCSTR lpCurrentDirectory:** This parameter is a pointer to a null-terminated string that specifies the current directory for the new process.

21. Write the naming convention for the client and server when they are on the same machine versus a different machine in Named Pipes?

ANSWER:

The client and server of the same machine

- \\.\pipe\[path]pipename

The client and server of the same machine

- \\servername\pipe\pipename

22. What are based pointers. Write their syntax.

ANSWER:

Base pointer is actually offset and related to the other pointer.

Syntax:

❖ Type__ based (base) declarator

23. Multi-threaded network, is it possible to attach same DLL file? Entry point in DLL file is Compulsory or optional?

ANSWER:

Yes, it is possible to attach the same DLL file in a multi-threaded network application. The entry point in a DLL file is optional.

24. If CreateProcess() returns successfully then it returns 2 handles, write their resources.

ANSWER:

When CreateProcess() returns successfully

It returns two handles:

- ❖ process handles
- ❖ thread handles.

The process handle is used to manipulate the process object, while the primary thread handle is used to manipulate the main thread of the newly created process.

25. What WaitForSingleObject() Returns if mutex is abandoned?

ANSWER:

WaitForSingleObject() is used to wait for a single specified object
If mutex is abandoned the return type is WAIT_ABANDONED

26. WaitForMultipleObjects

ANSWER:

WaitForMultipleObjects can be used for more than 1 object, those objects are defined in the form of an array.
return type is DWORD.

27. How many synchronization objects Windows Provide?

ANSWER:

Windows Provide four synchronization objects

- ❖ Mutexes
- ❖ Semaphores
- ❖ Events
- ❖ Critical Section

28. What is the return type of dwopenmutex

ANSWER:

The return type is a HANDLE.

29. Write the Syntax of InterlockedIncrement and InterlockedDecrement

ANSWER:

LONG InterlockedIncrement(

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```
  LONG volatile *lpAddend
);

LONG InterlockedDecrement(
  LONG volatile *lpAddend
);
```

30. Threadpool Ky parameter and return type.

ANSWER:

the Ky parameter in the Threadpool API is a void pointer that can be used to pass additional data to the callback function executed by the thread pool worker thread, and the return type of the callback function depends on the specific implementation and requirements of the application.

31. Return type of ThreadPool() is Boolean

ANSWER:

- CreateThreadpool()
- SubmitThreadpoolWork()
- WaitForThreadpoolWorkCallbacks().

32. WaitForThreadpoolWorkCallbacks() function explain

ANSWER:

The WaitForThreadpoolWorkCallbacks() function is used to wait for all callbacks for a specified work object to complete before continuing execution. which waits for all pending work associated with a thread pool to complete. The function takes a handle to a thread pool, and an optional flag to indicate whether to cancel any remaining work items before waiting.

33. How many threads can enter the critical section?

ANSWER:

Only one thread can enter a critical section at a time. If another thread tries to enter the same critical section while it is already occupied by another thread, it will be blocked until the first thread exits the critical section.

34. Briefly explain system mask and process mask.

ANSWER:

In Windows programming, the system mask and process mask are bitmaps used to indicate which processor(s) a thread is allowed to run on.

The **system mask** represents all available processors on the system, while the **process mask** represents the subset of processors that are available to a particular process.

35. Briefly describe each parameter in the following Widows API.

BOOL GetProcessTime(HANDLE hProcess,

LPFILETIME lpCreationTime,

LPFILETIME lpExitTime,

LPFILETIME lpKernelTime,

LPFILETIME lpUserTime);

ANSWER:

The GetProcessTimes() Windows API is used to retrieve timing information about a specified process. The timing information include the process that is creation time, exit time, kernel mode execution time and user mode execution time. It's return type is BOOL and it has Five parameters.

36. What is the purpose of dwPipeMode parameter in CreateNamedPipe() API and mention the names of its two commonly used values.

ANSWER:

dwPipeMode parameter in CreateNamedPipe() can affect the behavior of Writing to the Pipe.

The two commonly used values for this parameter are

- ❖ PIPE_WAIT
- ❖ PIPE_NOWAIT

37. Which Windows API is used to release the Ownership of mutex? Also Write its Signature

ANSWER:

The Windows API used to release the ownership of a mutex is called ReleaseMutex().

The signature of the ReleaseMutex() API is as follows:

BOOL ReleaseMutex(HANDLE hMutex);

38. Name any third-party frameworks that are used for thread optimization.

ANSWER:

- ❖ OpenMP
- ❖ cilk++

39. Synchronization mutex thread thread model.

ANSWER:

A synchronization mutex is used in thread synchronization to ensure that only one thread can access a critical section or a shared resource at a time. This is a key component of the thread synchronization model, which is used to coordinate and control the execution of multiple threads in a concurrent program.

OR

Synchronization refers to the coordination of multiple threads or processes

to ensure that they access shared resources in a mutually exclusive and coordinated manner. The **mutex model** is widely used in multithreaded programming to prevent race conditions and ensure data consistency in shared resources.

40. **Syntax allocate and deallocate function**

ANSWER:

Allocate:

allocate(<data_type>, <size>)

Deallocate:

deallocate(<pointer>)

41. Interlocked Function.

ANSWER:

Interlocked are mostly suited if the variables with volatile scopey only need to incremented, decremented and exchange. It is faster, simpler and easy to use.

The most basic of these functions are InterlockedIncrement() and InterlockedDecrement().

42. Types of callback Functions.

ANSWER:

- CreateThreadpoolTimer()
- CreateThreadpoolIO()
- CreateThreadpoolWait()

43. What is Critical section?

ANSWER:

Critical section is the part of the code that can only executed by one thread at a time. CRITICAL_SECTION objects are initialized and deleted but not a handle and not shared among process. Only one process can enter into the critical section. If another thread enters into the critical section while it has already occupied by another thread it will be blocked until when the first thread exit from the critical section.

44. Write the name of setting pointer.

ANSWER:

- SetFilePointer()
- SetFilePointerEx()

45. Write the types of floating-point exception?

ANSWER:

- Invalid operation
- Division by zero
- Overflow

- Underflow
- Inexact calculation

46. Write the name of API that is used to remove the lock from the locked file area?

ANSWER:

The name of API is used to remove the lock from the locked file area depends on the operating system and programming being used.

47. CreatePipe parameters API

ANSWER:

BOOL CreatePipe (
PHANDLE phRead,
PHANDLE phWrite,
LPSECURITY_ATTRIBUTES lpsa,
DWORD cbPipe);

48. SetThreadAffinityMask

ANSWER:

DWORD_PTR SetThreadAffinityMask (
HANDLE hThread,
DWORD dwThreadAffinityMask);

49. Write the GetProcessAffinityMask parameters API for get and set these masks a set of APIs

ANSWER:

BOOL GetProccessAffinityMask (
HANDLE hprocess,
LPDWORD lpProcessAffinityMask,
LPDWORD lpSystemAffinityMask);

50. SetProcessAffinityMask()

ANSWER:

BOOL SetProcessAffinityMask (
HANDLE hNamedPipe,
DWORD_PTR dwProcessAffinityMask);

51. InterlockedExchangeAdd write the parameter or return type

ANSWEWR:

LONG InterlockedExchangeAdd(
LONG volatile *Addend,
LONG increment)

return type and parameters is Loong Integer

52. When thread Object is created, the space stack memory?

ANSWER:

Yes, when thread object is created a space in memory is allocated for

Thread's Stack. The stack is a portion of memory that is used to store local variables, function parameters, and return addresses for the functions called by the thread.

53. Write the name of functions or APIs used for the following tasks.

I. Creating a job object
II. Opening a named job object
III. Adding a process to a job object
IV. To destroy or close job object

ANSWER:

I. The function used for creating a job object is **CreateJobObject**.

II. The function used for opening a named job object is **OpenJobObject**.

III. The function used for adding a process to a job object is **AssignProcessToJobObject**.

IV. The function used for destroying or closing a job object is **CloseHandle**.

54. Mention the names of two type of events and which function is used to set these types?

ANSWER:

Two types of events are

- Manual-reset events
- Auto-reset events.

A manual-reset event can signal several waiting threads simultaneously and can be reset. An auto-reset event signals a single waiting thread, and the event is reset automatically.

55. What is the name of Windows API that is used to get and report error to the user?

ANSWER:

The Windows API that is commonly used to get and report errors to the user is called getlasterror() and FormatMessage

56. Write the name of three windows functions that are used to manage Mutexes?

ANSWER:

- CreateMutex()

- ReleaseMutex()
- OpenMutex()

57. Write the general syntax of APIs thar are used for:

I. Allocating index in the Thread Local Storage (TLS)
II. Deallocating index in the Thread Local Storage (TLS)

ANSWER:

I. Allocating index in the Thread Local Storage (TLS)
DWORD TlsAlloc(VOID);
II. Deallocating index in the Thread Local Storage (TLS)
BOOL TlsFree(DWORD dwTlsIndex);

58. Write the Functionalities of the Following APIs.

I. GetLastError()
II. FormatMessage()
III. LocalFree()

ANSWER:

- **GetLastError()**: It will be used to get Last error.
- **FormatMessage():** retrieves the system error message associated with a given error code and formats it into a user-readable string.

OR:

It translate the error code into a meaningful Languag

- **LocalFree() frees**: It is also used with **formateMessage()** to deallocate the allocate memory

59. In the CreateNamePipe() API function the dwOpenMode specifies manay flags. Name any Three

ANSWER:

- PIPE_ACCESS_DUPLEX
- PIPE_ACCESS_INBOUNT
- PIPE_ACCESS_OUTBOUND

60. Keeping in View the following syntax of DuplicationHandle() API, answer the questions.

BOOL DuplicateHandle (
HANDLE hSourceProcessHandle,
HANDLE hSourceHandle,
HANDLE hTargetProcessHandle,

LPHANDLE lpTargetHandle,
DWORD dwDesiredAccess,
BOOL bInheritHandle,
DWORD dwOptions);

I. **Mention the name of parameter that shows the original or real handle.**

OR

Which argument will provide the original source handle?

II. **Mention the name of parameter that receives a copy of the original handle.**

OR

Which target handle will receive the pointer of original source?

NOTE: HERE is TWO type Question can be asked

ANSWER:

I. hSourceHandle
II. lpTargetHandle

61. Consider the Following Code given below for using Thrtead pools.

```
Int_tmain(DWORD argc,LPTSTR argv[]){
INT nThread;
HANDLE *pWorkobjects;
THARG *pThreadArg;
TP_CALLBACK_ENVIRON cbe;
Nthread-_ttoi(argv[]);
```

You need to write code to first allocate memory to pWorkobjects using maloc() then use CreateThreadpoolWork() api function to create a thread pool worker object.
Considering the name of work call back is "Worker".
ANSWER:

```
// Allocate memory for the array of work objects
pWorkobjects = (HANDLE*)malloc(nThread * sizeof(HANDLE));
```

```
// Initialize the thread pool callback environment
InitializeThreadpoolEnvironment(&cbe);

// Create the thread pool worker objects
for (int i = 0; i < nThread; i++) {
   pThreadArg = (THARG*)malloc(sizeof(THARG));
   pThreadArg->worknum = i;
   pWorkobjects[i] = CreateThreadpoolWork(Worker, pThreadArg, &cbe);
}
```

62. Pulseevent its return type and parameters

ANSWER:

BOOL PulseEvent (HANDLE hEvent)

Return type is BOOL

63. SetEvent its return type and parameters

ANSWER:

BOOL SetEvenet (HANDLE hEvent)

Return type is BOOL

64. ResetEvent its return type and parameters

ANSWER:

BOOL ResetEvent (HANDLE hEvent)

Return type is BOOL

65. Write the APIs Parameter of Openfilemapping.

ANSWER:

```
HANDLE OpenFileMapping (
DWORD dwDesiredAccess,
BOOL bInheritHandle,
LPCTSTR lpMapName
);
```

return type is Handle

66. Explain Memory barrier

ANSWER:

A memory is access in desired order use memory barrier. The interlocked functions provide memory barrier. A Memory barrier is a Hardware or software mechanism used memory is access in specific order. It

is used Synchronize data access between Multiple threads and prevent unexpected result caused by inconsistent data ordering.

67. While CreatingNamedPipe, which options are available in dwOpenMode

ANSWER:

dwOpenMode is the open mode which can be.

- PIPE_ACCESS_DUPLEX,
- PIPE_ACCESS_INBOUNT,
- PIPE_ACCESS_OUTBOUND along with many other options.

68.In Create file mapping, use these characteristics and show the API

- Given file name is hIN
- You do not want to set any security attributes.
- You want to use the file size.
- The file should be only PAGE_READONLY
- You do not want to share the map with any other process.

69.How many Simple Job management Command Line?

ANSWER:

There are three Simple Job Management Command Line.

- Jobs
- Jobbg
- Kill

70.What is set functions?

ANSWER:

Set functions typically refer to a group of system calls or functions that allow processes to manipulate sets of file descriptors or other types of system resources.

Some common set functions in operating systems include

- Select()
- waitpid()
- sigprocmask()

71.Write the Functionality of flags?

ANSWER:

Flags are variables or values that can be set or cleared to indicate a specific condition or status within a program or system. They are commonly

used to control program flow, enable or disable features, and represent binary states. The main functionality of flags is to provide a simple and efficient way for programs to store and check the status of various conditions or settings. They allow programs to make decisions based on specific conditions or states, which can improve efficiency and accuracy.

72. Write the parameter of DLLMain().

ANSWER:

```
BOOL DllMain
(
        HINSTANCE hDll,
        DWORD Reason,
        LPVOID lpReserved
)
```

73. Is this statement correct fiber are scheduled by OS and thread are scheduled by application? Is this correct or incorrect.

ANSWER:

It is incorrect statement,
Because Fiber schedule by Application and thread Schedule By OS

74. CreateThreadpoolWork

ANSWER:

```
PTP_WORK CreateThreadpoolWork (
PTP_WORK_CALLBACK pfnwk,
PVOID pv,
PTP_CALLBACK_ENVIRON pcbe);
```

75. As a default, if a system Affinity mask=24 bits, then What is indicate?

ANSWER:

24-bit affinity mask in a system indicates that the mask can represent up to 24 processors or processor groups

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